Kartikey Pathak

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Education

Veermata Jijabai Technological Institute (VJTI) Mumbai, India
B.Tech in Electrical Engineering, Minor in Robotics (CGPA: 8.04) *2023 – 2027*

Experience

Eyecandy Robotics April 2026 – August 2026
Electronics and Hardware Engineer *Bangalore, India*

- Designed production PCBs for a legged consumer robot: Raspberry Pi Zero 2W and Milk-V compute carriers, 1S/2S battery charging and power boards, and servo driver boards
- Owned schematic capture, routing and fabrication output, integrating IMU, servo bus, audio and microphone array with test points for hardware bring-up

NishCorp Technologies July 2025 – November 2025
Robotics Control Systems Engineer Intern *Hybrid*

- Developed Gazebo simulation for a robotic wheelchair platform to analyze motion and steering, applying differential drive kinematics and integrating IMU plugins
- Contributed to the half-scale hardware prototype through integration, testing and field evaluation

Krishna Defence and Allied Industries Limited April 2025 – July 2025
Research Intern - Defence Technology *IIT Gandhinagar*

- Developed firmware for Qorvo DWM3001CDK Ultra-Wideband modules using Zephyr RTOS for indoor positioning where GPS is unreliable, handling low-level drivers and RF packets
- Implemented SLAM and NAV2 for autonomous forklift POC navigation in warehouse environments
- Achieved precision localization error of 30 cm across a 10,000 sq meter area and 2 cm in a 50 sq meter area

Projects

PCB Axial-Flux BLDC Motor | *KiCAD, FEMM, Python* Aug 2026 – Present

- Designing a coreless axial-flux motor whose stator is the PCB itself: 100 mm two-layer board, 6 wedge coils over 8 poles, dual rotor with 16 N42 magnets across a 0.5 mm air gap
- Built FEMM magnetostatic models to extract torque constant, back-EMF and torque ripple, identifying a 3.3x torque loss from the plastic rotor before fabrication

FOC Controller for Micro BLDC Motor | *STM32, DRV8311H, PCB Design* Jan 2026 – Feb 2026

- Developed a Field-Oriented Control (FOC) system for a 4300 KV micro BLDC motor operating from a 3.7 V supply
- Designed and routed a custom PCB integrating an STM32 microcontroller, DRV8311H motor driver, and AS5600 encoder

Smart Switch Board | *ESP32, ESP-IDF, MQTT, BLE, Flutter, PCB Design* 2025 – Present

- Built a network-controlled mains wall switch across every layer: CAD enclosure, a relay and optocoupler board isolating 220 V from a 3.3 V ESP32, ESP-IDF firmware, a TLS MQTT broker and a Flutter app
- Kept the physical switches working with network, broker and app unavailable, using BLE WiFi provisioning and state persisted in NVS across power cuts

Open Manipulator X Simulation | *ROS2, Gazebo, RViz, Computer Vision* June 2024 – July 2024

- Programmed forward and inverse kinematics for an Open Manipulator X robot arm to perform precise pick-and-place tasks
- Implemented object detection and tracking algorithms to locate target objects in the Gazebo simulation environment

Competitions

eYantra Warehouse Drone Competition | *IIT Bombay* Oct 2024 – Feb 2025

- Developed autonomous navigation and PID control algorithms for stable hovering and motion of a simulated drone, with ArUco marker detection for localization and task execution

Positions of Responsibility

Society of Robotics and Automation (SRA) Aug 2024 – Mar 2026
Mechanical Head & Active Member *VJTI, Mumbai*

- Conducted technical workshops on Embedded C, ESP32 microcontrollers, and ROS-based systems for 200+ students
- Mentored junior teams in developing self-balancing and line-following robots, providing hardware/software expertise

Technical Skills

Programming Languages: C, C++, Python, Embedded C
Software & Frameworks: ROS2, Gazebo, RViz, ESP-IDF, STM32CubeIDE, KiCAD, micro-ROS
Hardware: ESP32, STM32, Raspberry Pi, UWB Modules, Motor Drivers, BLDC/Stepper/DC Motors, LiDAR
Core Competencies: Robot Kinematics, Field-Oriented Control, SLAM, NAV2, PID Control, PCB Design, IoT